

SENTINEL

Slide blueprint for the pitch deck

HOW TO USE THIS	One page per slide. Build the slide from the left column, put the numbers in verbatim, say the line at the bottom.
LENGTH	12 slides, about 8 minutes. Timings are on each page.
RULE	Every number here comes from the repo. Re-run <code>`python -m src.report`</code> before the deck is final.
LAST PAGE	Four things not to say. Read it before you rehearse.

ALL DATA SIMULATED — SAY SO ON SLIDE 1

Title

WHAT GOES ON THE SLIDE

- Project name + one-line definition
- Team names
- The words "All data simulated" — visible, not hidden
- Keep it plain. No stock imagery.

THE NUMBERS — USE THESE EXACTLY

Deck length	8 min
Slides	12

SAY THIS

“SENTINEL is an explainable screening layer for component burn-in. Everything you'll see runs on simulated data, and I'll show you why that's a strength.”

The stake

WHAT GOES ON THE SLIDE

- One sentence of story, large type
- A satellite image OR nothing at all
- NO numbers on this slide — it is the only emotional one you get

THE NUMBERS — USE THESE EXACTLY

Mission	7 years
Failed at	14 months
Repairs possible	0

SAY THIS

“A satellite power controller fails 14 months into a seven-year mission. Root cause: one part with a resistive via that passed every screening test on the line. There is no maintenance window in orbit.”

The problem, quantified

WHAT GOES ON THE SLIDE

- ONE number, as large as the slide allows
- Sub-line underneath in small type
- This is your strongest slide — do not clutter it

THE NUMBERS — USE THESE EXACTLY

Latent defects passing every datasheet limit **174 / 174**

Recall of static limits **0.000**

Parts static limits DO flag **55 – all gross failures**

SAY THIS

“Every single latent defect in our dataset passes every datasheet limit at 168 hours. Static screening catches none of them. That is the gap.”

Why static limits miss it

WHAT GOES ON THE SLIDE

- The drift chart — lot envelope + one escaping part
- Screenshot it from the console, Part Detail screen
- Annotate: 'legal at every read point'

THE NUMBERS — USE THESE EXACTLY	
Part L04-0009 at 0h	34.5 nA
At 168h	123.5 nA
Lot median at 168h	26.8 nA
Datasheet limit	200 nA

SAY THIS

“This part never touches the 200 nanoamp limit — it is legal at every read point. It is also drifting at four times its lot's rate. Static limits ask whether a part is legal. They never ask whether it is behaving.”

Architecture

WHAT GOES ON THE SLIDE

- One diagram, left to right
- Module A → Module B → Fusion → reason codes
- Label the three non-negotiables underneath

THE NUMBERS — USE THESE EXACTLY

Module A	per-lot robust stats
Module B	forecast from hour 24
Fusion	0-100, three bands

SAY THIS

“Three things make this work. Currents are lognormal, so we log-transform before any statistic. We use median and 1.4826 times MAD, never mean and sigma — a mean-based limit gets masked by the very outliers it is hunting. And every statistic is computed within a lot.”

Module A — detection

WHAT GOES ON THE SLIDE

- A four-row table
- Highlight the shipped row
- One footnote about reproducing the reference

THE NUMBERS — USE THESE EXACTLY

Static limits	recall 0.000
L2 Dynamic PAT	PR-AUC 0.3889 · R@10% 0.839
L3 pooled evidence ← shipped	PR-AUC 0.4083 · R@10% 0.874
MinCovDet (comparison)	PR-AUC 0.4038

SAY THIS

“Our L2 layer reproduces the published reference implementation to three decimal places. That is what tells you the feature layer underneath everything else is correct.”

Module B — forecast from hour 24

WHAT GOES ON THE SLIDE

- MAE table: parameter, n*, ours, linear, last-value
- Colour the winner in each row
- Then the honest caveat, and the money line

THE NUMBERS — USE THESE EXACTLY

Beats linear extrapolation by	43-67%
Iddq_uA MAE	1.29 vs 2.35 linear
n* on Tpd / Vol	0.00 — no extrapolable signal
Oven hours freed	14,976

SAY THIS

“The power law beats linear extrapolation by 43 to 67 percent. But on timing and voltage the fitted exponent is zero — the data supports no extrapolation at all there, and plain last-value-carried-forward wins. We report that rather than hide it.”

The decision policy

WHAT GOES ON THE SLIDE

- Three band cards: ACCEPT / WATCH / REJECT
- Lot table underneath with L04 highlighted
- This is the demo moment — show it live if you can

THE NUMBERS — USE THESE EXACTLY

ACCEPT / WATCH / REJECT	1,785 / 210 / 105
Recall, REJECT+WATCH	0.810 at 6.0% overkill
Recall at a 5% budget	0.776
L04 reject rate	9.7% vs L03's 2.3%

SAY THIS

“REJECT is sized to the five percent Percent Defective Allowable, so the screen cannot scrap a good lot. And L04 comes out at 9.7 percent against L03's 2.3 — that is the deliberately bad lot, found without ever being told it exists.”

Explainability

WHAT GOES ON THE SLIDE

- One real reason code, quoted in full
- The one-page PDF report beside it
- End on: "this is what the inspector signs"

THE NUMBERS — USE THESE EXACTLY

Reason codes emitted	682 across 285 parts
Codes	R-101 ... R-601
Signed PDFs generated	105

SAY THIS

“Every flag carries the part’s actual value and its lot reference value, in datasheet units, in a sentence. These come from rules, not from a model — which is why they are auditable. This is what the QA inspector signs.”

What we withdrew

WHAT GOES ON THE SLIDE

- Four rows: status chip + claim + one line
- Use CORRECTED / REFUSED / MEASURED as the chips
- THIS IS YOUR DIFFERENTIATOR — no other team has this slide

THE NUMBERS — USE THESE EXACTLY

No correlation break	rho 0.153 vs null 0.189
Accuracy	refused – scorer raises an error
PR-AUC tie artefact	0.09 'gain' withdrawn
Reason code R-301	fired on 25.8% – recalibrated

SAY THIS

“Each of these made one of our numbers worse. They are on the slide because a judge who checks will find them, and because they are the reason you can trust the numbers that are left.”

The ceiling is metrology, not modelling

WHAT GOES ON THE SLIDE

- Recall split by carrying parameter
- Then the sensitivity table
- This turns your weakest number into your strongest claim

THE NUMBERS — USE THESE EXACTLY

Current-carried defects	recall 0.79 - 0.87
Timing/voltage-carried	recall 0.24 - 0.42
At 1.5% tester noise	0.65
At 0.4% tester noise	0.92 — model unchanged

SAY THIS

“Where we are blind is not where our model is weak — it is where the tester cannot measure. Improve timing repeatability from 1.5 percent to 0.4 and the same unchanged pipeline reaches 0.92. A better model cannot recover signal that was never measured.”

Close

WHAT GOES ON THE SLIDE

- The closing line, large
- Reproducibility strip underneath
- QR code or short URL to the live console

THE NUMBERS — USE THESE EXACTLY

Tests passing	150
Seed	42, byte-identical
Setup	one command: <code>bash run_demo.sh</code>

SAY THIS

“Static limits ask whether a part is legal. SENTINEL asks whether it is behaving like its siblings, and where it will be in a week. In a system you cannot repair, that is the difference between a screen and a gamble.”

Four things not to say

1. Never show accuracy

About 8% of parts are defective, so predicting "all good" scores 92% and catches nothing. If a judge asks why it is missing, that sentence IS the answer — and it earns you credibility for free.

2. Never say "correlation break"

The delta-vector covariance in this dataset is diagonal. A judge who checks will find it. Say "it pools marginal evidence across parameters" instead — which is what it actually does, and is still a real gain.

3. Do not claim Module B is as accurate as the full screen

At hour 24 it catches roughly a third of latent defects, against about 87% at hour 168. It is a triage layer that buys 144 oven-hours per part pulled. That trade stands on its own — overclaiming it is the first thing a reliability engineer will probe.

4. Do not hide that the data is simulated

Say it on slide 1. A well-argued synthetic dataset lets you prove recall against known ground truth, which real unlabelled burn-in data never can. It is a strength, not an apology.

If a judge pushes on any of these, the honest answer is the strong answer. You measured it, you wrote it down, and you changed what you claimed.